Abstract Algebra II

Homework 2

Due: 18th March 2026

Submit only Questions 1 and 2. (All fields are assumed to have **characteristic 0**.)

Note. For this exercise sheet it would be helpful to recall the First Sylow theorem (by yourself). This is as much Sylow theory as you need for this course.

1. Let $K/\mathbb{C}$ be a field extension of degree $2^a m$, for some integer $a \geq 0$, and some odd integer $m \geq 1$.

   (a) Show that $m = 1$.

   (b) Show that $a = 0$.

2. We say that a field extension L/K is *normal* if the minimal polynomial over K of every element in L splits in L, i.e., every zero of that polynomial also lies in L.

   (a) Show that $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not a normal extension.

   (b) Show that $\mathbb{Q}\left(\sqrt[4]{2}, \zeta_8\right)/\mathbb{Q}$ is normal and compute the degree of the extension. Here ζ_8 denotes the primitive 8th-root of unity.

3. Fix a prime p. Let K be a field such that for any field extension L/K, we have

   $$1 < [L : K] < \infty \implies p \mid [L : K].$$

   Prove that every finite extension of K has degree p^m for some $m \geq 0$.

4. A field extension L/K is said to be *Galois* if it is normal and separable, here the latter condition is automatic in characteristic zero. Suppose L/K is a Galois extension such that $[L : K] = p^m$ for some prime p and integer $m \geq 1$. Prove that there exist $m-1$ intermediate fields $K_1, ..., K_{m-1}$ such that we have a chain of inclusions

   $$K = K_0 \subsetneq K_1 \subsetneq K_2 \subsetneq \cdots \subsetneq K_{m-1} \subsetneq K_m = L$$

   satisfying

   (a) $[K_i : K_{i-1}] = p$, and

   (b) K_i/K is Galois,

   for all $1 \leq i \leq m$.